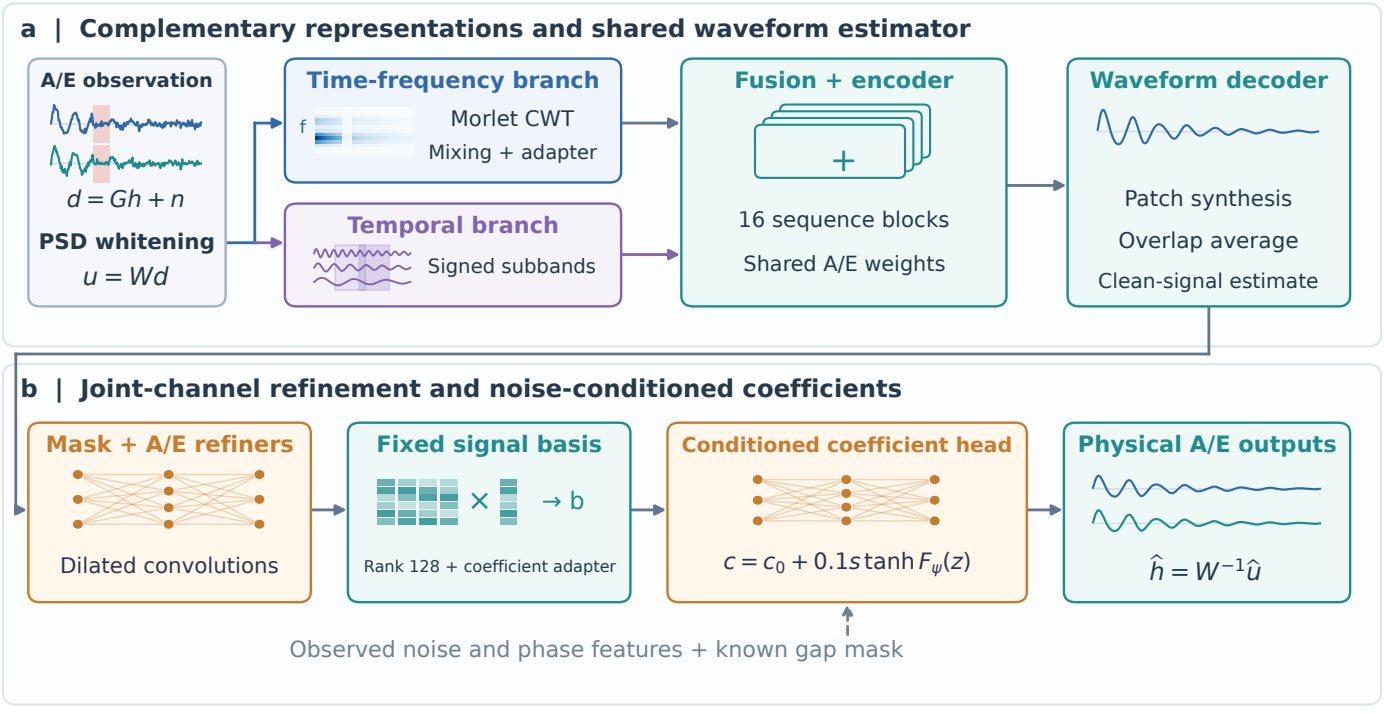

a | Complementary representations and shared waveform estimator
A/E observation
$d = Gh + n$
PSD whitening
$u = Wd$
Time-frequency branch
f
Morlet CWT
Mixing + adapter
Temporal branch
Signed subbands
Fusion + encoder
+
16 sequence blocks
Shared A/E weights
Waveform decoder
Patch synthesis
Overlap average
Clean-signal estimate
b | Joint-channel refinement and noise-conditioned coefficients
Mask + A/E refiners
Dilated convolutions
Fixed signal basis
× → b
Rank 128 + coefficient adapter
Conditioned coefficient head
$c = c_0 + 0.1s\tanh F_\psi(z)$
Physical A/E outputs
$\hat{h} = W^{-1}\hat{u}$
Observed noise and phase features + known gap mask